GEOMETRY ON ONE PAGE

From polygons to the circle

Inscribed polygons squeeze a circle from inside, circumscribed polygons from outside, and doubling the sides closes the gap.

1 / THE RULE AND THE PICTURE

A circle of radius 1 lies between its inscribed and circumscribed regular polygons



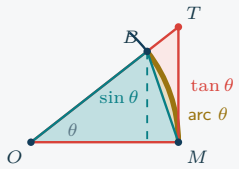
$n = 4$ $5.657 < 2\pi < 8.000$ | $n = 8$ $6.123 < 2\pi < 6.627$ | $n = 16$ $6.243 < 2\pi < 6.365$ | $n = 32$ $6.273 < 2\pi < 6.303$

The inscribed polygon (green) has its vertices on the circle and the circumscribed polygon (red) has its sides touching it, so the circle's circumference 2π is trapped between the two perimeters p_n and P_n, and the trap narrows as the number of sides n doubles:

$$p_n = 2n \sin\tfrac{\pi}{n} \;<\; 2\pi \;<\; 2n \tan\tfrac{\pi}{n} = P_n$$

2 / THE PROOF OF THE BOUNDS / ONE TRIANGLE AND ONE SECTOR

Comparing three areas shows $\sin\theta < \theta < \tan\theta$ for the half-angle $\theta = \pi/n$



Take one side of the polygon, seen from the centre O as half of the central angle $2\pi/n$. The green triangle OMB has area $\frac{1}{2}\sin\theta$, the gold sector has area $\frac{1}{2}\theta$, and the red triangle OMT has area $\frac{1}{2}\tan\theta$, and each region contains the one before it, so

$$\tfrac{1}{2}\sin\theta < \tfrac{1}{2}\theta < \tfrac{1}{2}\tan\theta.$$

Multiplying by $2n$ and using $n\theta = \pi$ turns the three areas into $p_n/2 < \pi < P_n/2$: the inscribed side is shorter than its arc, which is shorter than the tangent segment.

3 / THE PROGRESSION / DOUBLING WITH SQUARE ROOTS ONLY

Start with the hexagon, $p_6 = 6$ and $P_6 = 4\sqrt{3}$, and double the sides with two means

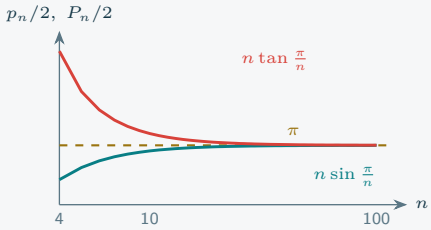
$$P_{2n} = \frac{2\,p_n P_n}{p_n + P_n} \text{ (harmonic mean)}, \qquad p_{2n} = \sqrt{p_n P_{2n}} \text{ (geometric mean)}$$

n	p_n	P_n	$p_n/2 < \pi < P_n/2$	gap $P_n - p_n$	gap ratio
6	6.00000	6.92820	$3.00000 < \pi < 3.46410$	0.9282	
12	6.21166	6.43078	$3.10583 < \pi < 3.21539$	0.2191	4.24
24	6.26526	6.31932	$3.13263 < \pi < 3.15966$	0.0541	4.05
48	6.27870	6.29217	$3.13935 < \pi < 3.14609$	0.0135	4.01
96	6.28206	6.28543	$3.14103 < \pi < 3.14271$	0.0034	4.00

The means reproduce $4n\tan\frac{\theta}{2}$ and $4n\sin\frac{\theta}{2}$ exactly, and at $n = 96$ the bounds sit inside Archimedes' $3\frac{10}{71} < \pi < 3\frac{1}{7}$, that is $3.1408 < \pi < 3.1429$. Each doubling cuts the gap by a factor of about 4.

4 / THE SQUEEZE / THE CIRCLE AS THE LIMITING CASE

Both perimeters tend to 2π, so the polygon with infinitely many sides is the circle



Writing $\theta = \pi/n$ we have $p_n = 2\pi \cdot \dfrac{\sin\theta}{\theta}$ and $P_n = 2\pi \cdot \dfrac{\tan\theta}{\theta}$, and as $n \to \infty$ the angle $\theta \to 0$ and both ratios tend to 1, so both perimeters tend to 2π. The ratio $P_n/p_n = 1/\cos\frac{\pi}{n}$ also tends to 1. The areas behave the same way, the inscribed area $\frac{n}{2}\sin\frac{2\pi}{n}$ and the circumscribed area $n\tan\frac{\pi}{n}$ both tending to π (giving $A = \pi r^2$), with 3.1394 and 3.1427 at $n = 96$.

5 / WHAT THE PROOF SHOWS / EXHAUSTION

A circle is never a polygon, and it is the one shape that every polygon sequence approaches

The argument needs only two facts, that a straight segment is the shortest path between two points and that the bounds close in on a single number, which is the definition of π. The perimeter $p_n = n\,|1 - \zeta_n|$ is n times the chord to the next root of unity, so the inscribed polygons here are those of the cyclotomic sheet, with arcs $\frac{1}{n}$ of the circle.

ASIDE / MEANS AGAIN

The doubling step uses a harmonic mean and then a geometric mean, the same two means H and G that measured the shapes of boxes.

ASIDE / CHECK IN PYTHON

`2*96*sin(pi/96)` gives 6.28206 and `2*96*tan(pi/96)` gives 6.28543.

Convention. Circle of radius 1, circumference 2π; p_n and P_n are perimeters of inscribed and circumscribed regular n-gons. After Archimedes' method of exhaustion.

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